gpeyt1-07 · Time rate of consolidation

GEO PE · Long Calculation Problems I — Topic 2: Soil Mechanics, Lab Testing & Analysis

Problem

An 18 ft layer of soft clay lies between a sand blanket above and a fractured, free-draining sandstone below. A preload is placed instantly. Lab data: cv = 0.28 ft²/day, Cc = 0.38, e0 = 1.12. Ultimate primary settlement under the preload is estimated at 9.6 in. The contractor wants to know how long the preload must stay to reach 90% average consolidation.

Most nearly, the time for U = 90% is:

A. 981 days
B. 57 days
C. 27 days
D. 245 days

Given equations / data

> t = Tv Hd²/cv; Hd = longest drainage distance; Tv(U = 90%) = 0.848, Tv(50%) = 0.197

Worked solution

1. Double drainage (sand top, free-draining rock base) → Hd = H/2 = 9 ft
2. t90 = 0.848 × 9² / 0.28 = 245 days (≈ 0.67 yr)
3. Settlement at that time ≈ 0.9 × 9.6 = 8.6 in. (extra data not needed)

Final answer

245 days

Why the other values are wrong

- **981 days** — used Hd = full thickness (single drainage)
- **57 days** — used Tv for 50%
- **27 days** — forgot to square Hd

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.2.3 Time Rate of Settlement and U–Tv table, p. 85

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=IGS94AWmPWA (no creator values copied).

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